

3	(a) $4.5 \times 50 - 2.8 \times M$ (= ...)	C1
	$(...) = -1.8 \times 50 + 1.4 \times M$	C1
	$(M =) 75 \text{ g}$	A1 [3]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2015	9702	21

(b) total initial kinetic energy/KE not equal to the total final kinetic energy/KE

or relative speed of approach is not equal to relative speed of separation

so not elastic or is inelastic

B1 [1]

(c) force on X is equal and opposite to force on Y (Newton III)

M1

force equals/is proportional to rate of change of momentum (Newton II)

M1

time of collision same for both balls hence change in momentum is the same

A1 [3]